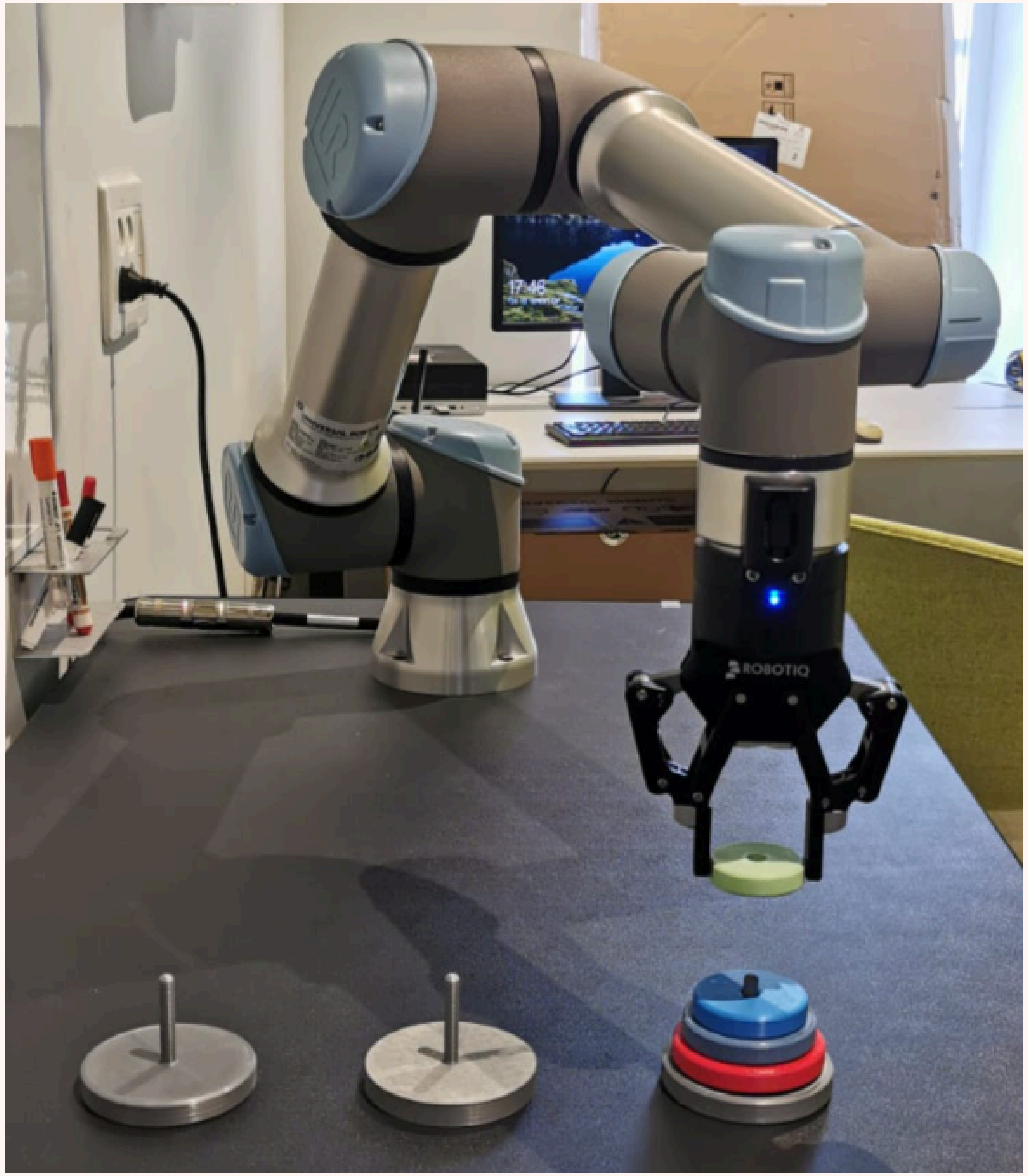
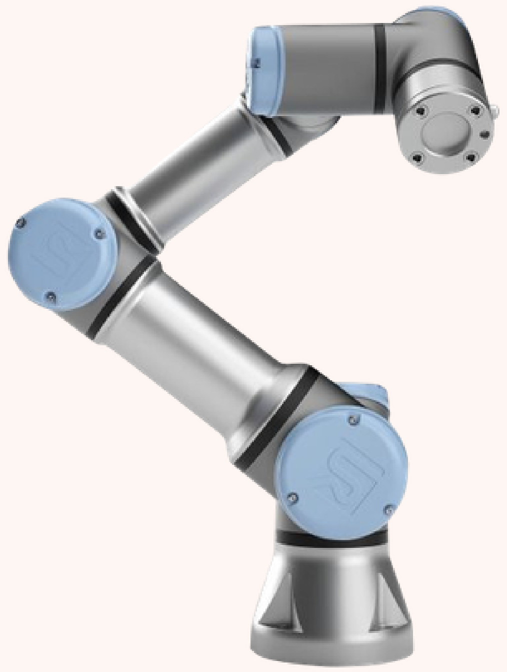
SIckKinematiX



Tomek Stolarczyk

Robot



UR5

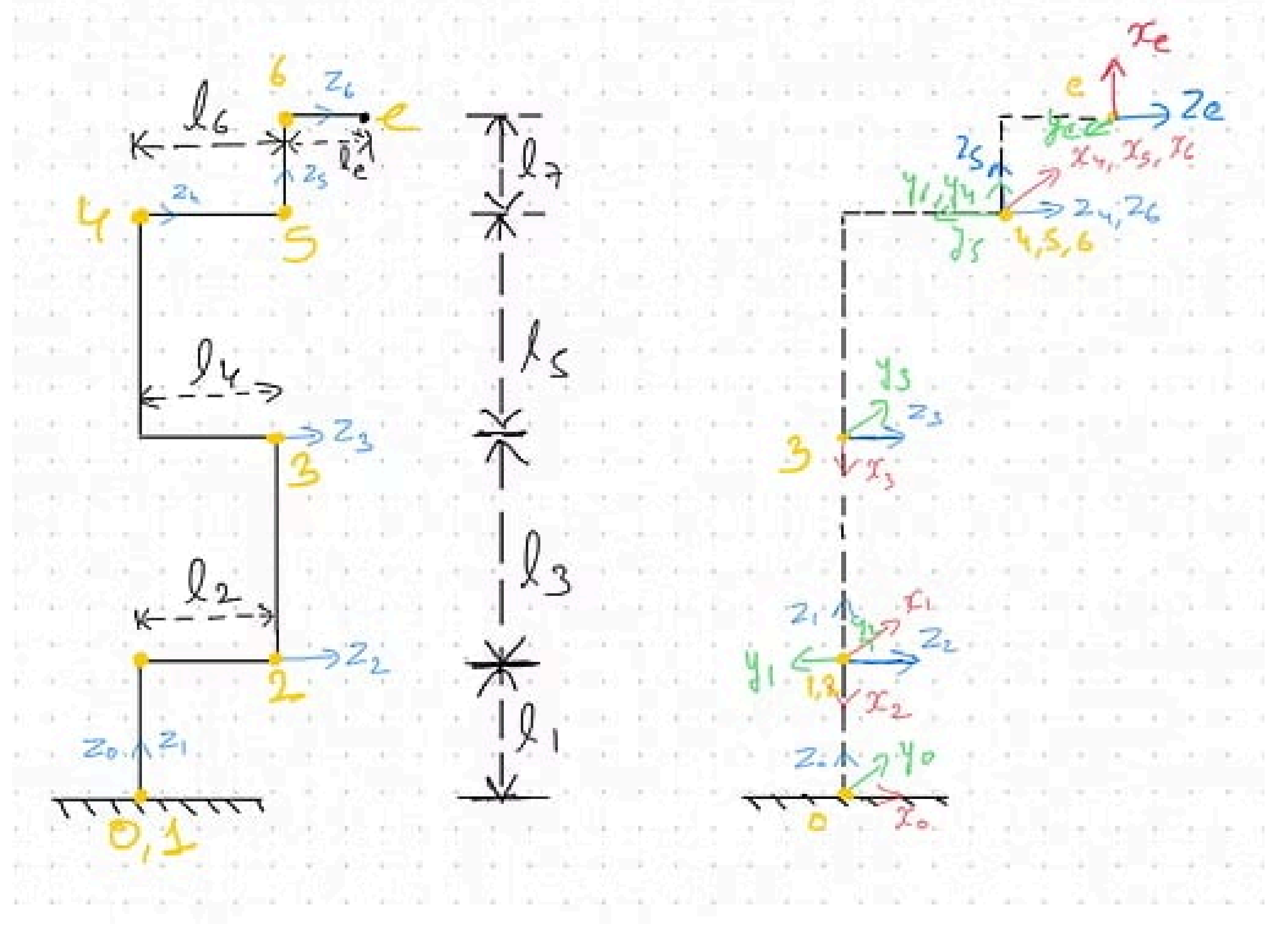
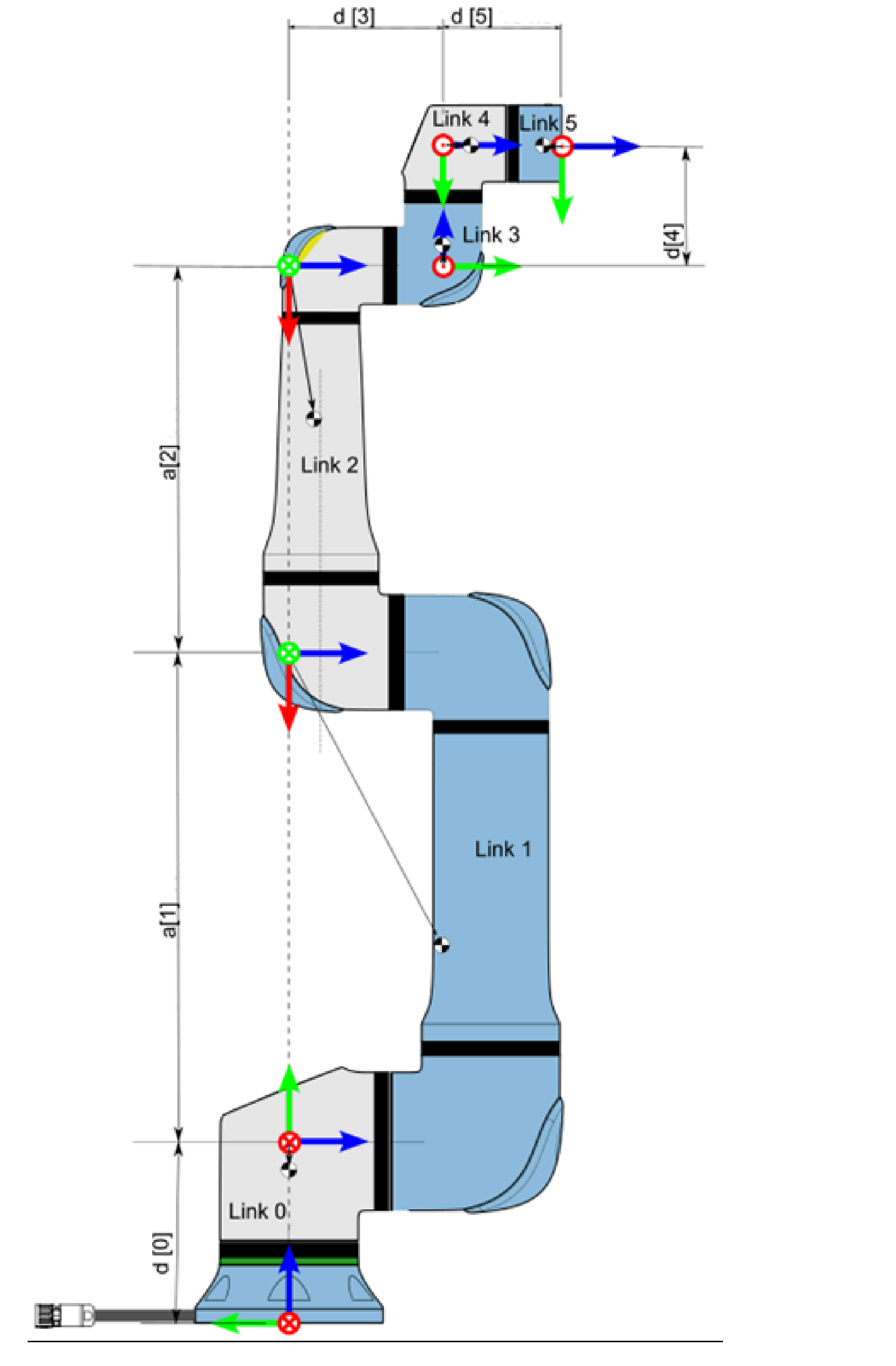
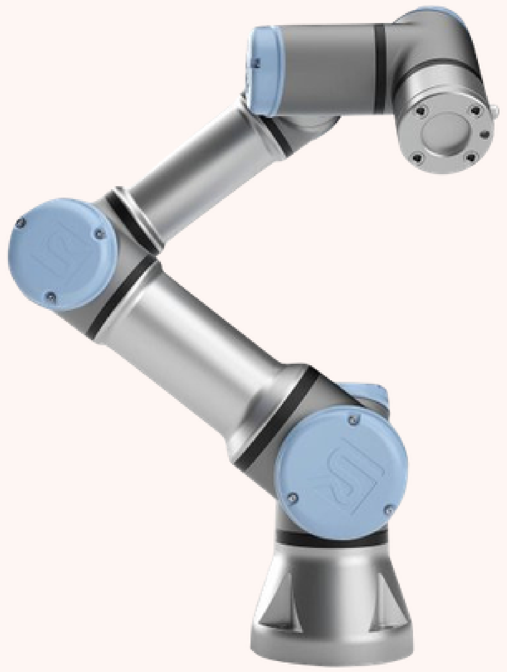


Fig. 2: Actual Joints(left) and Modified DH Frames(right) of 6-axis Manipulator



UR5						
Kinematics	a [m]	d [m]	alpha [rad]	**Dynamics**	Mass [kg]	Center of Mass [m]
Joint 1	0	0.089159	π/2	Link 1	3.7	[0, -0.02561, 0.00193]
Joint 2	-0.425	0	0	Link 2	8.393	[0.2125, 0, 0.11336]
Joint 3	-0.39225	0	0	Link 3	2.33	[0.15, 0.0, 0.0265]
Joint 4	0	0.10915	π/2	Link 4	1.219	[0, -0.0018, 0.01634]
Joint 5	0	0.09465	-π/2	Link 5	1.219	[0, 0.0018,0.01634]
Joint 6	0	0.0823	0	Link 6	0.1879	[0, 0, -0.001159]



UR5

Robot

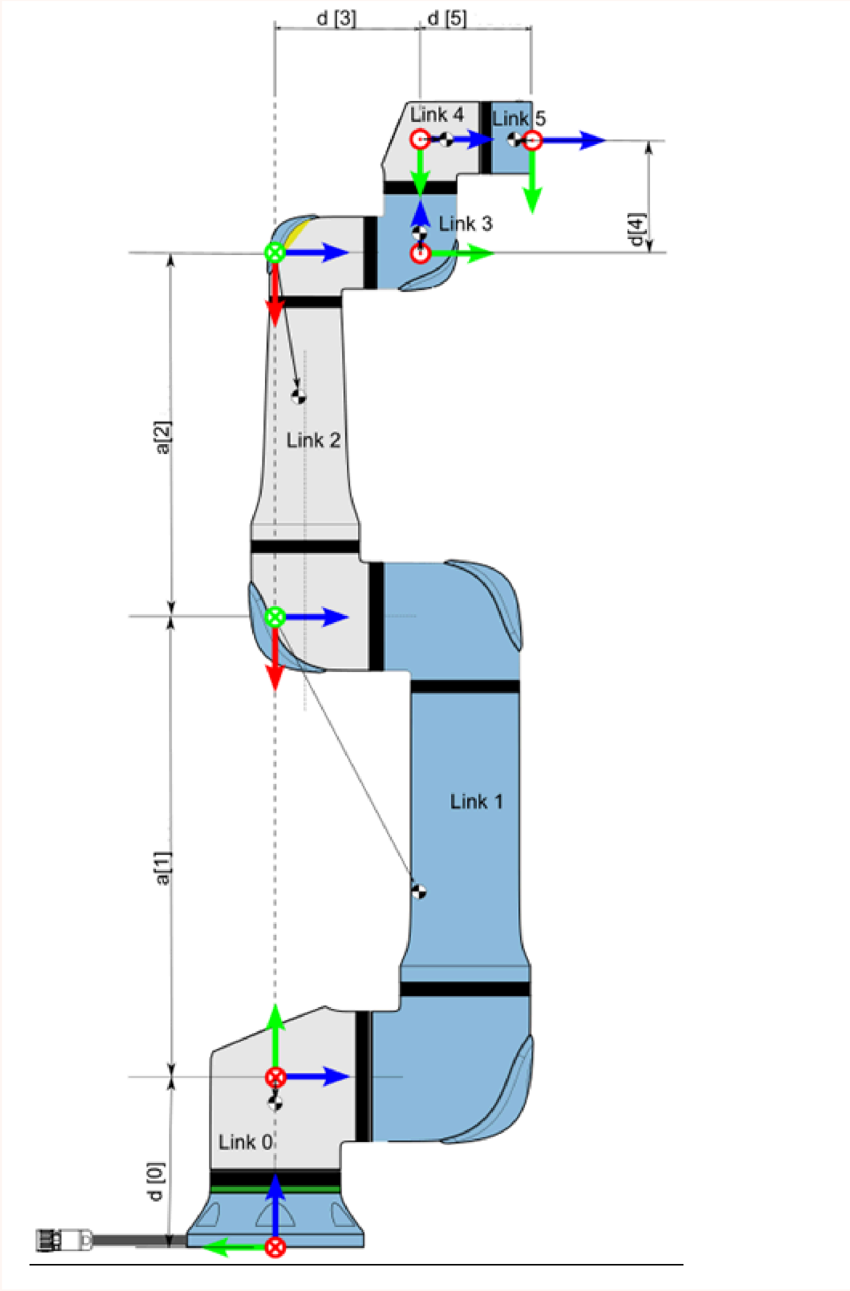
Modified DH Parameters for 6 DOF Arm

Joint (i)	α_{i-1} (rad)	a_{i-1} (m)	d_i (m)	θ_i (rad)
1	0	0	l_1	$\theta_1 + \pi/2$
2	$\pi/2$	0	0	$\theta_2 - \pi/2$
3	0	$-l_3$	0	θ_3
4	0	$-l_5$	l_6	$\theta_4 + \pi/2$
5	$-\pi/2$	0	0	θ_5
6	$\pi/2$	0	0	$\theta_6 + \pi/2$

$$T_{i-1}^{i} = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} & 0 & a_{i-1} \\ s_{\theta_i} c_{\alpha_{i-1}} & c_{\theta_i} c_{\alpha_{i-1}} & -s_{\alpha_{i-1}} & -d_i s_{\alpha_{i-1}} \\ s_{\theta_i} s_{\alpha_{i-1}} & c_{\theta_i} s_{\alpha_{i-1}} & c_{\alpha_{i-1}} & d_i c_{\alpha_{i-1}} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

where:

- **cos(θ_i)** and **sin(θ_i)** represent the rotation around the z_i axis.
- **cos(α_i)** and **sin(α_i)** represent the twist between the z_{i-1} and z_i axes.
- $\mathbf{a_i}$ is the distance between the z_{i-1} and z_i axes along the x_i axis.
- $\mathbf{d_i}$ is the distance along the z_i axis from the origin of the i-th coordinate frame to the x_i axis.



Gdzie jest Robot?

Czyli Kinematyka Prosta

a) Transformation between Frame 1 and Base Frame

$$T_{10} = \begin{bmatrix} -\sin(\theta_1) & 0 & -\cos(\theta_1) & 0 \\ \cos(\theta_1) & 0 & -\sin(\theta_1) & 0 \\ 0 & 1 & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

b) Transformation between Frame 2 and Base Frame

$$T_{20} = T_{21} \cdot T_{10}$$

$$T_{20} = \begin{bmatrix} -\sin(\theta_1)\sin(\theta_2) & -\sin(\theta_1)\cos(\theta_2) & \cos(\theta_1) & 0 \\ \sin(\theta_2)\cos(\theta_1) & \cos(\theta_1)\cos(\theta_2) & \sin(\theta_1) & 0 \\ -\cos(\theta_2) & \sin(\theta_2) & 0 & l \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

c) Transformation between Frame 3 and Base Frame

$$T_{30} = T_{32} \cdot T_{21} \cdot T_{10}$$

$$T_{30} = \begin{bmatrix} -\sin(\theta_1)\sin(\theta_2+\theta_3) & -\sin(\theta_1)\cos(\theta_2+\theta_3) & \cos(\theta_1) & l_3\sin(\theta_1)\sin(\theta \\ \sin(\theta_2+\theta_3)\cos(\theta_1) & \cos(\theta_1)\cos(\theta_2+\theta_3) & \sin(\theta_1) & -l_3\sin(\theta_2)\cos(\\ -\cos(\theta_2+\theta_3) & \sin(\theta_2+\theta_3) & 0 & l_1+l_3\cos(\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

d) Transformation between Frame 4 and Base Frame

$$T_{40} = T_{43} \cdot T_{32} \cdot T_{21} \cdot T_{10}$$

$$T_{40} = \begin{bmatrix} -\sin(\theta_1)\sin(\theta_2+\theta_3) & -\sin(\theta_1)\cos(\theta_2+\theta_3) & \cos(\theta_1) & l_3\sin(\theta_1)\sin(\theta \\ \sin(\theta_2+\theta_3)\cos(\theta_1) & \cos(\theta_1)\cos(\theta_2+\theta_3) & \sin(\theta_1) & -l_3\sin(\theta_2)\cos(\\ -\cos(\theta_2+\theta_3) & \sin(\theta_2+\theta_3) & 0 & l_1+l_3\cos(\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

e) Transformation between Frame 5 and Base Frame

$$T_{50} = T_{54} \cdot T_{43} \cdot T_{32} \cdot T_{21} \cdot T_{10}$$

$$T_{50} = \begin{bmatrix} -\sin(\theta_1)\cos(\theta_2+\theta_3+\theta_4) & \sin(\theta_1)\sin(\theta_2+\theta_3+\theta_4) & \cos(\theta_1) & l_3\sin(\theta_1)\sin(\theta_2)+l_5\sin \\ \cos(\theta_1)\cos(\theta_2+\theta_3+\theta_4) & -\sin(\theta_2+\theta_3+\theta_4)\cos(\theta_1) & \sin(\theta_1) & -l_3\sin(\theta_2)\cos(\theta_1)-l_5 \\ \sin(\theta_2+\theta_3+\theta_4) & \cos(\theta_2+\theta_3+\theta_4) & 0 & l_1+l_3\cos(\theta \\ 0 & 0 & 0 & \end{bmatrix}$$

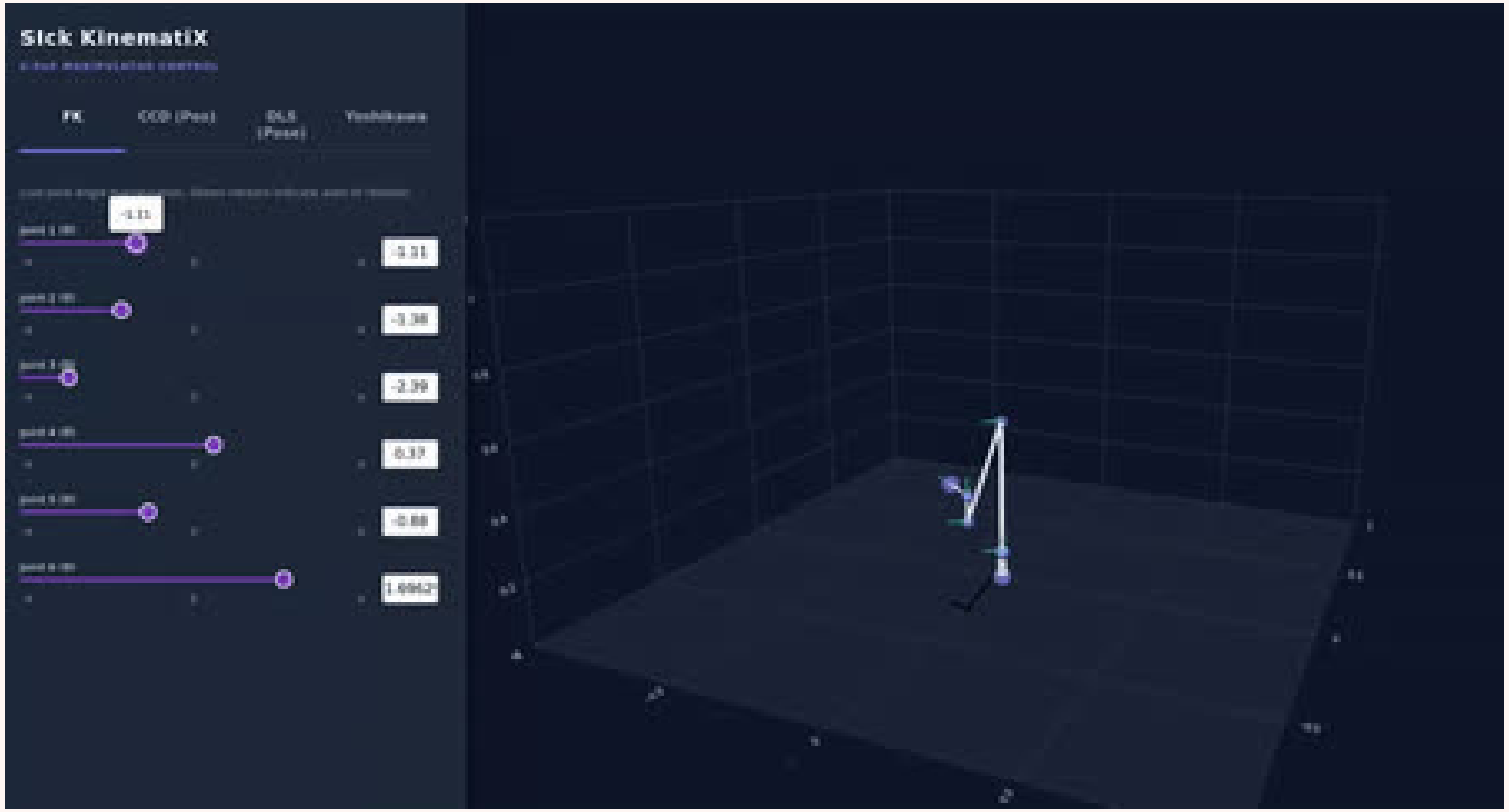
f) Transformation between Frame 6 and Base Frame

$$T_{60} = T_{65} \cdot T_{54} \cdot T_{43} \cdot T_{32} \cdot T_{21} \cdot T_{10}$$

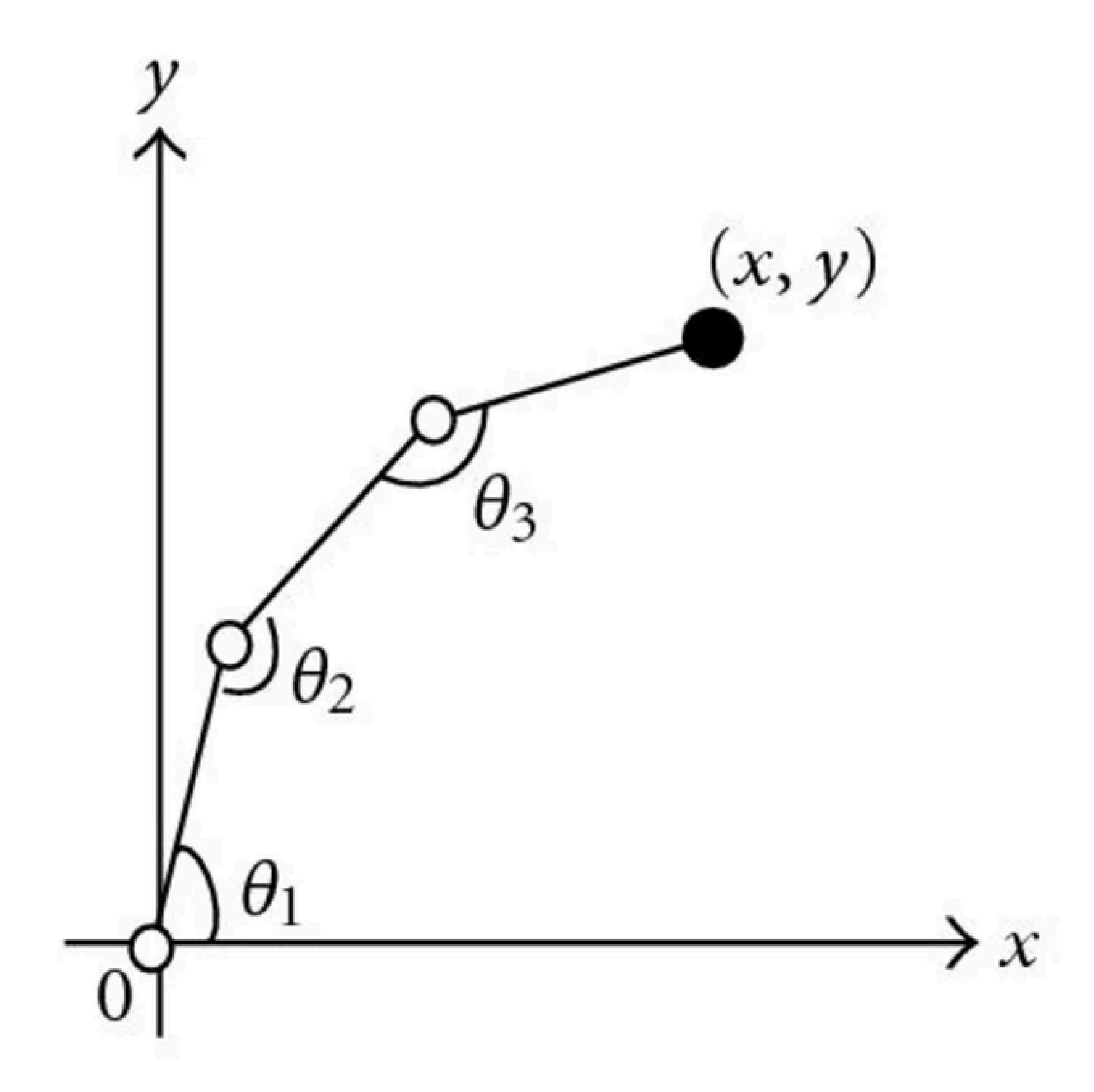
$$T_{60} = \begin{bmatrix} (\sin(\theta_1)\cos(\theta_5)\cos(\theta_2+\theta_3+\theta_4)+\sin(\theta_5)\cos(\theta_1))\sin(\theta_6)+\sin(\theta_1)\sin(\theta_2+\theta_3+\theta_4)\cos \\ (\sin(\theta_1)\cos(\theta_5)\cos(\theta_2+\theta_3+\theta_4)+\sin(\theta_5)\cos(\theta_1))\cos(\theta_6)-\sin(\theta_1)\sin(\theta_6)\sin(\theta_2+\theta_3 \\ -\sin(\theta_1)\sin(\theta_5)\cos(\theta_2+\theta_3+\theta_4)+\cos(\theta_1)\cos(\theta_5) \\ l_3\sin(\theta_1)\sin(\theta_2)+l_5\sin(\theta_1)\sin(\theta_2+\theta_3)+l_6\cos(\theta_1) \end{bmatrix}$$

4x4 matrix math

Kinematyka Prosta



Kinematyka Odwrotna - algorytm CCD



3d vector math

Kinematyka Odwrotna - Jakobian i algorytm DLS

$$\begin{bmatrix}\Delta X\\ \Delta Y\\ \Delta Z\\ \Delta q_i\\ \Delta q_j\\ \Delta q_k\end{bmatrix}=\begin{bmatrix}\frac{\partial X}{\partial\theta_1} & \frac{\partial X}{\partial\theta_2} & \frac{\partial X}{\partial\theta_3} & \frac{\partial X}{\partial\theta_4} & \frac{\partial X}{\partial\theta_5} & \frac{\partial X}{\partial\theta_6}\\ \frac{\partial Y}{\partial\theta_1} & \frac{\partial Y}{\partial\theta_2} & \frac{\partial Y}{\partial\theta_3} & \frac{\partial Y}{\partial\theta_4} & \frac{\partial Y}{\partial\theta_5} & \frac{\partial Y}{\partial\theta_6}\\ \frac{\partial Z}{\partial\theta_1} & \frac{\partial Z}{\partial\theta_2} & \frac{\partial Z}{\partial\theta_3} & \frac{\partial Z}{\partial\theta_4} & \frac{\partial Z}{\partial\theta_5} & \frac{\partial Z}{\partial\theta_6}\\ \frac{\partial q_i}{\partial\theta_1} & \frac{\partial q_i}{\partial\theta_2} & \frac{\partial q_i}{\partial\theta_3} & \frac{\partial q_i}{\partial\theta_4} & \frac{\partial q_i}{\partial\theta_5} & \frac{\partial q_i}{\partial\theta_6}\\ \frac{\partial q_j}{\partial\theta_1} & \frac{\partial q_j}{\partial\theta_2} & \frac{\partial q_j}{\partial\theta_3} & \frac{\partial q_j}{\partial\theta_4} & \frac{\partial q_j}{\partial\theta_5} & \frac{\partial q_j}{\partial\theta_6}\\ \frac{\partial q_k}{\partial\theta_1} & \frac{\partial q_k}{\partial\theta_2} & \frac{\partial q_k}{\partial\theta_3} & \frac{\partial q_k}{\partial\theta_4} & \frac{\partial q_k}{\partial\theta_5} & \frac{\partial q_k}{\partial\theta_6}\end{bmatrix}\cdot\begin{bmatrix}\theta_1\\ \theta_2\\ \theta_3\\ \theta_4\\ \theta_5\\ \theta_6\end{bmatrix}$$

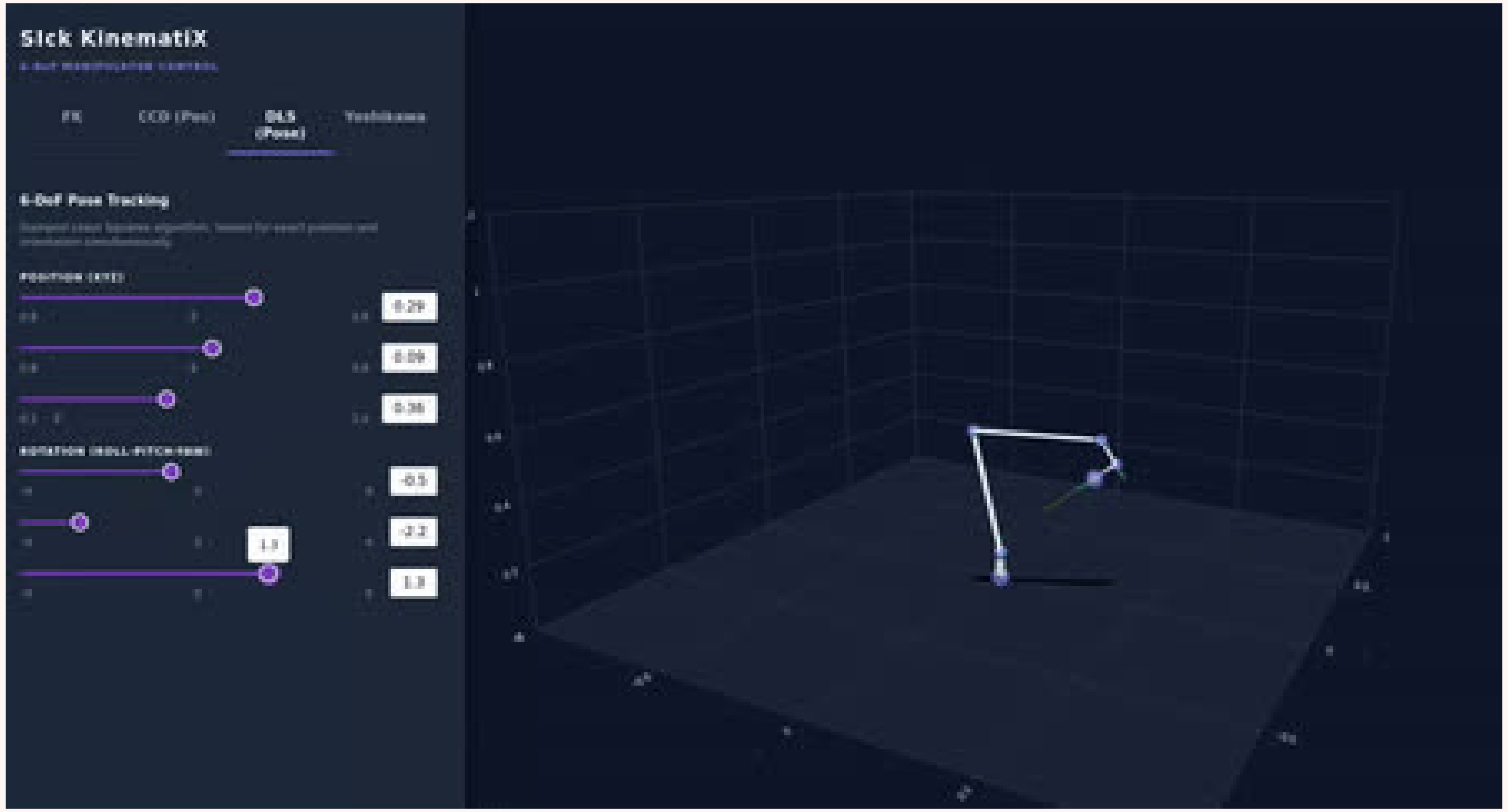
$$\dot{q}=\mathbf{J}(q)^{-1}\nu$$

$$\Delta\theta=(J^TJ+\lambda I)^{-1}J^T\Delta x$$

6x6 vector math

finding inverse matrix as well as det with gaussian elimination

Kinematyka Odwrotna - algorytm DLS



teraz obejrzymy to w akcji...

Workspace & Index Yoshikawa

$$w = \sqrt{\det(JJ^T)}$$

Monte Carlo

teraz obejrzymy to w akcji 2...